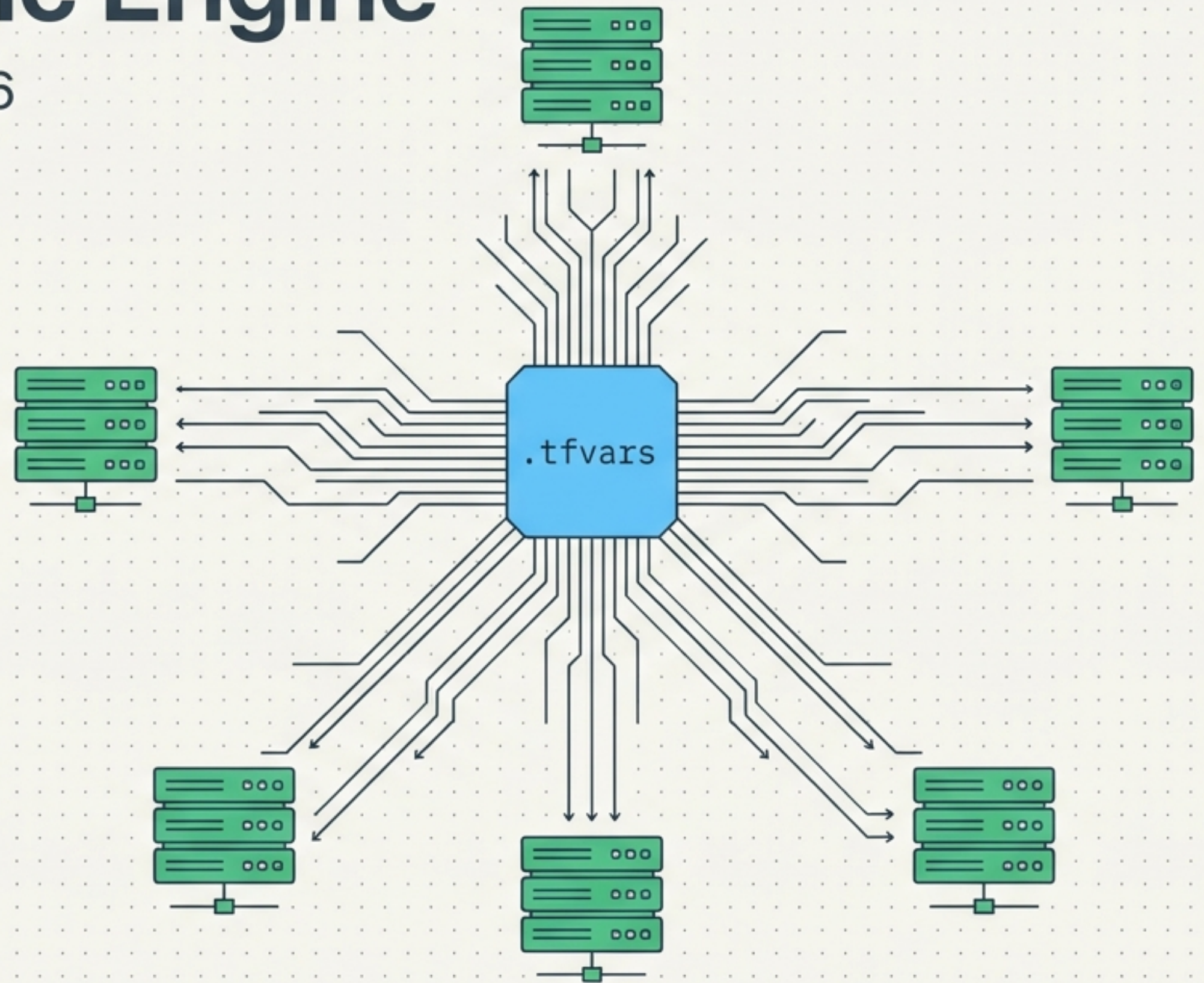


# The Deterministic Engine

Master Control Dials for Vision 2026  
Azure Cloud Infrastructure



TARGET: AZURE HUB-SPOKE

FRAMEWORK: ZERO-TRUST / 100% IAC

SCALE: 259 RESOURCES | 170 PIPELINES



# IaC Sovereignty & The DRY Logic Engine

100% of infrastructure is managed as software.

Zero configuration drift allowed.

Manual Azure Portal changes are automatically overwritten.

Raw  
Variables

**locals.tf**  
The Instance Generator

Deterministic  
Environment

High-level architectural intent is automatically translated into hundreds of low-level Azure resources without duplicating a single line of HCL code.



# The 3-Tier Configuration Hierarchy Matrix



**The Blueprint**  
`variables.tf`

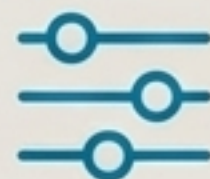
**Purpose:** Defines structure, types, and global defaults.

---

**Persistence:** Static, version-controlled.

---

**Sensitivity:** Public/Safe.



**The Environment Dials**  
`.tfvars`

**Purpose:** Injects target environment flags, SKUs, and topology switches.

---

**Persistence:** Dynamic per pipeline execution.

---

**Sensitivity:** Sanitized/Reference only.



**The Vault**  
Azure Key Vault

**Purpose:** Holds OBO tokens, TLS certs, and database passwords.

---

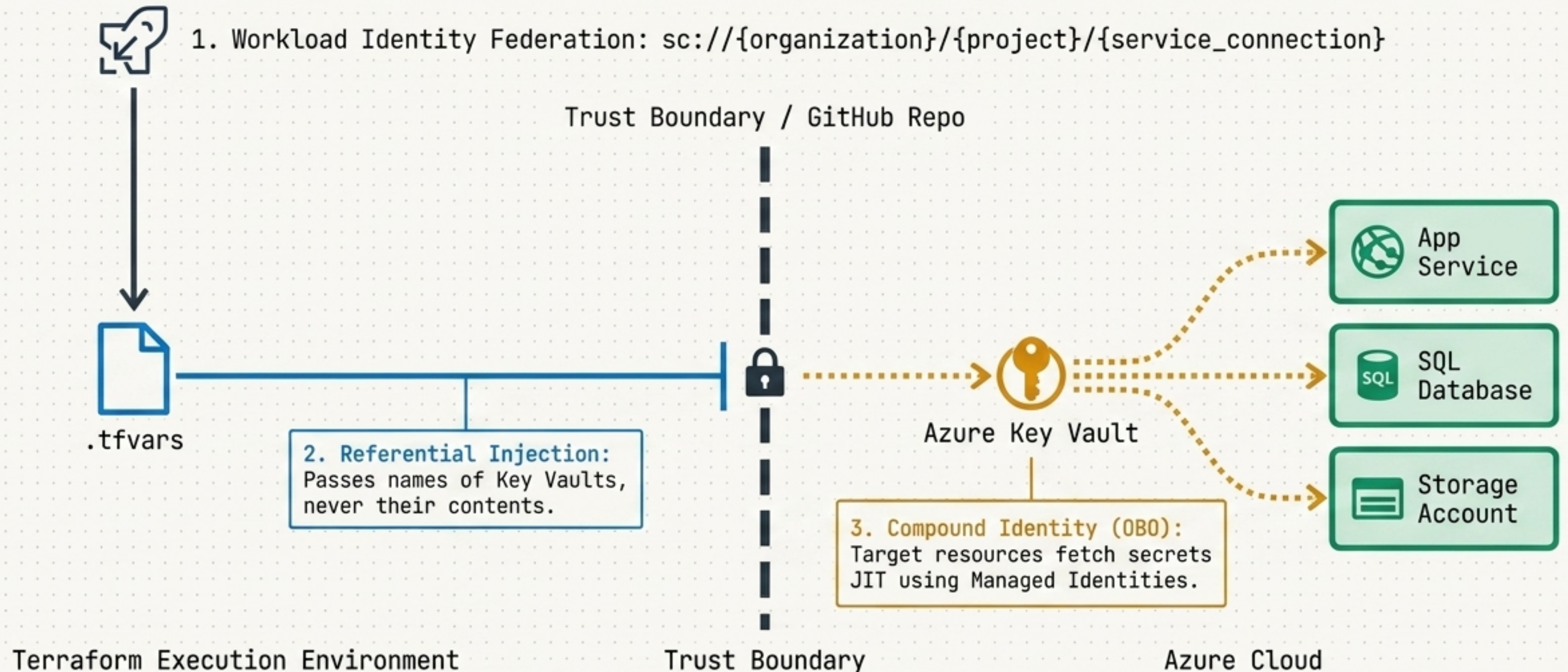
**Persistence:** Ephemeral, fetched JIT via identity.

---

**Sensitivity:** Strictly classified.



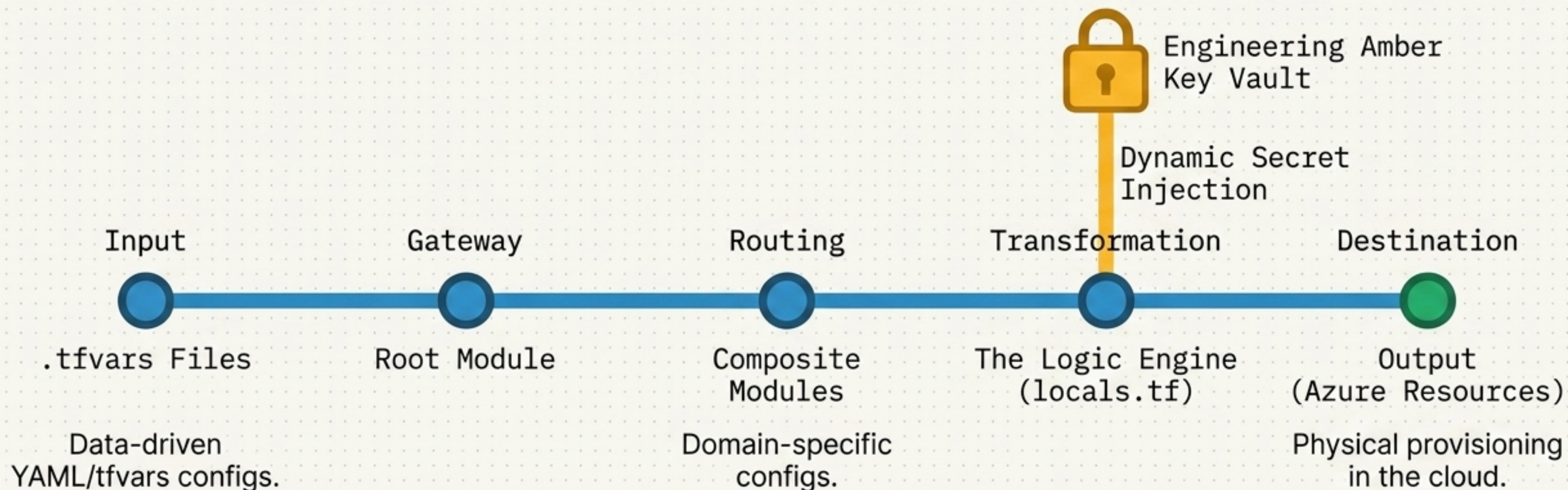
# The Zero-Secret Trust Boundary



`use_oidc = true` enforced across all core providers.  
Long-lived Service Principal secrets are completely eliminated.



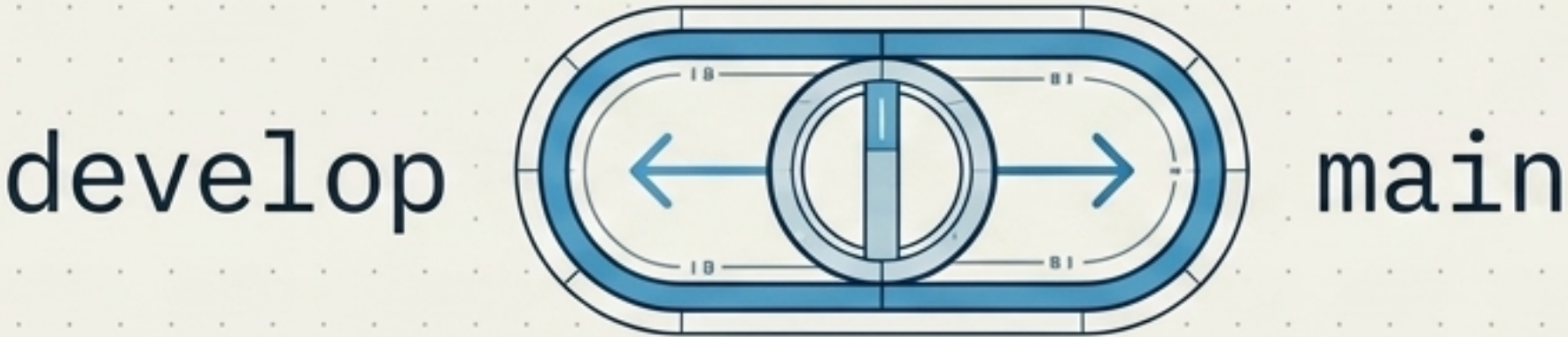
# Variable Flow Architecture



Execution Principle:  
Variables define the what;  
the Engine dictates the how.



# Branch Awareness: The Shadow Infrastructure Logic



## develop branch

|                |                                                                                     |                                                                           |
|----------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Code Snippet   | <code>{}</code>                                                                     | <code>gitbranch != "main" ? "d":""</code>                                 |
| Effect         |  | Prepends 'd' prefix to all resources.                                     |
| Resulting Name |  | <b>stdnedev</b> (Engineering Storage)                                     |
| Purpose        |                                                                                     | Shadow Infrastructure. IaC Lab Mode to test logic without collision risk. |

## main branch

|                |                                                                                       |                                                                |
|----------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Code Snippet   | <code>{}</code>                                                                       | <code>gitbranch == "main"</code>                               |
| Effect         |  | Clean naming, zero prefixes.                                   |
| Resulting Name |  | <b>stnepro</b> (Production Storage)                            |
| Purpose        |                                                                                       | Production sovereignty; the official state of the environment. |

A single Git branch switch instantly recalculates the entire global namespace, eliminating naming collisions.



# Tier Parity Strategy: ENG vs. PRO

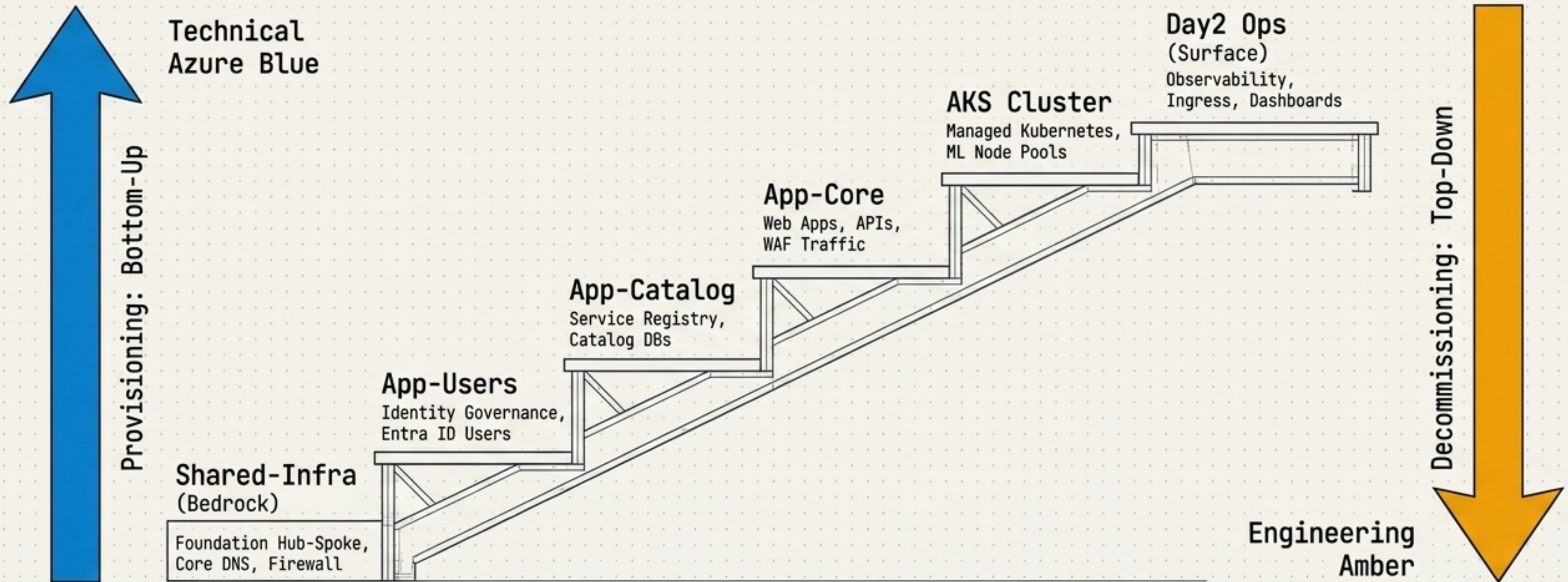
Identical HCL Code. Deterministic Variations.

|                  | Engineering (ENG) | Production (PRO)               |
|------------------|-------------------|--------------------------------|
| App Service SKU  | B1 or P1v2        | P2v2 or P3v3                   |
| AKS Node Pool    | Standard_DS2_v2   | Standard_D4s_v3 (Auto-healing) |
| MongoDB Tier     | M10 Standard      | M30+ High-Availability         |
| Data Replication | LRS / Local       | GRS / ZRS / Geo-Zone           |
| Backup Policy    | 7 Days Retention  | 180+ Days Retention            |

Strict Tier Parity ensures code tested in ENG behaves identically in PRO.  
The only variables are Scale and Cost.



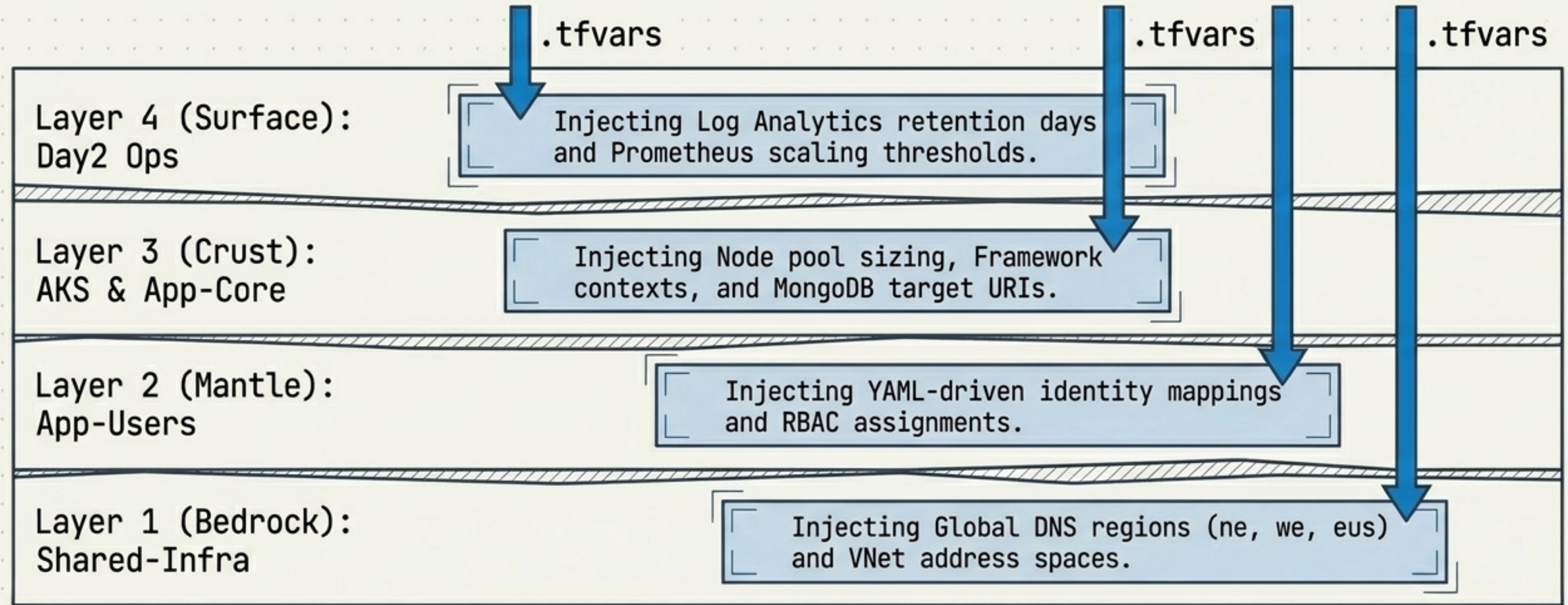
# The Reversible Dependency Staircase



Terraform State Locking guarantees isolation. Infrastructure must be instantiated base-first to satisfy VNet and Identity dependencies, and destroyed top-first to prevent orphan resources.



# Topographical Variable Injection



Referential Attribute Linking ensures upper layers dynamically consume IDs (`.id`, `.name`) generated by the bedrock layers.



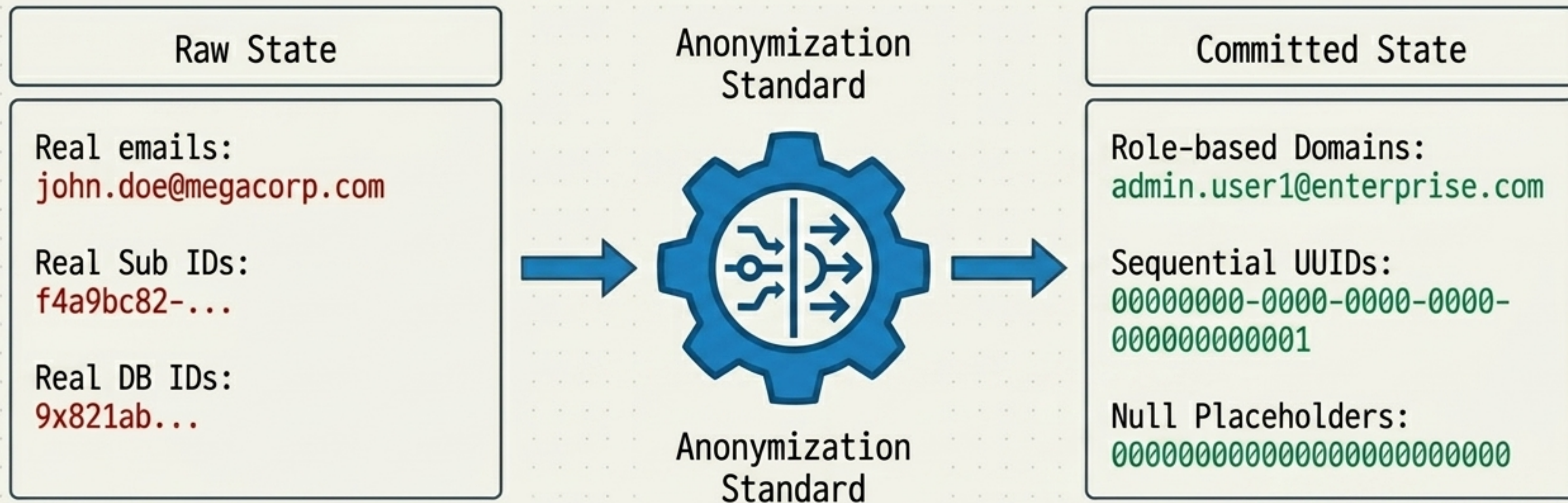
# The Sprawling Environment Matrix

|          |                                   | develop Branch (Shadow/Test) | main Branch (Stable/Prod) |
|----------|-----------------------------------|------------------------------|---------------------------|
| ENG Tier | DEV<br>(Rapid prototyping)        | dev-developbranch.tfvars     | dev-mainbranch.tfvars     |
|          | QA<br>(Stable WAF testing)        | qa-developbranch.tfvars      | qa-mainbranch.tfvars      |
|          | UAT<br>(End-to-End dummy data)    |                              |                           |
|          | PRE<br>(Pre-production staging)   |                              |                           |
| PRO Tier | PRO<br>(Live Core Production)     | pro-developbranch.tfvars     | pro-mainbranch.tfvars     |
|          | DEMO<br>(Specialized Public Demo) | dem-developbranch.tfvars     | dem-mainbranch.tfvars     |
|          | RES<br>(Researcher/Sandbox)       |                              |                           |
|          |                                   |                              |                           |

Experimental spaces (RES, DEMO) receive Production-grade security (Firewall, WAF) by being mapped to the PRO Shared-Infra tier.



# The Anonymization Filter: AI-Ready Configuration



Data must be sanitized at the code level. This guarantees that LLMs and AI coding assistants can ingest the entire repository context without exposing enterprise secrets.



# Synthesis: The Deterministic Engine



True IaC Sovereignty isn't about writing infrastructure. It is about parameterizing it so perfectly that a single text file can summon a massive, multi-tenant ecosystem on demand.